Srinivasa Sundara Rajan R
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EDUCATION

Master of Science in Microbiology
Kristu Jayanti College, Bengaluru, India

RESEARCH INTEREST

My research focuses on intersection of clinical microbiology and computational methods, particularly in understanding antimicrobial resistance mechanisms and developing biomarkers for bacterial infection. I am especially interested in leveraging bioinformatics and whole-genome sequencing to predict resistance patterns and improve early detection of hospital-acquired infections in vulnerable patient populations.

RESEARCH EXPERIENCE

SENIOR RESEARCH FELLOW | Nov 2023 - Present
National Institute of Mental Health and Neurosciences (NIMHANS), Bangalore

TECHNICAL ASSISTANT | May 2022 – Nov-2023
ICMR- National Institute of Virology, Bangalore

PROJECT TECHNICAL SUPPORT IV | September 2020 – May 2022
ICMR- National Institute of Virology, Bangalore

RESEARCH ASSISTANT | June 2020 – August 2020
Department of Life sciences, Kristu Jayanti College (Autonomous)

PROJECT ASSISTANT – II June 2018 – June 2019
CSIR-Central Electrochemical Research Institute, Karaikudi, India

PUBLICATIONS

Vidhya, T., **Rajan, R.S.S.** and Priya, J.S., 2026. Machine learning in diagnostic stewardship: A technical approach in improving diagnostic accuracy, optimizing antibiotic use in false positive CSF cultures. *Journal of Microbiological Methods*, p.107545.

Radhakrishnan, S.S.R., Bahubali, V.K.H., Birija, G.J.S. et al. Diagnostic Accuracy of Cerebrospinal Fluid Presepsin vs. Procalcitonin in Post-Neurosurgical Bacterial Ventriculitis/ Meningitis: A Machine Learning Analytical Approach. J Mol Neurosci 76, 18 (2026). https://doi.org/10.1007/s12031-026-02483-3

Gupta, S., **Rajan, R.S.S.**, Vidhya, T. *et al.* Evaluating the diagnostic significance of biomarkers in ventriculitis through a multivariate correlation study in a tertiary care center. *Sci Rep* **15**, 33754 (2025). https://doi.org/10.1038/s41598-025-97744-3

Rajan, R.S.S. (2024). Copper surface acts as good surface for biofilm attachment. *Asia Pacific Journal of Molecular Biology and Biotechnology*, 32(2), 31–37. https://doi.org/10.35118/apjmbb.2024.032.2.04

Rajan, R.S.S., Thomas, J., Francis, D., Daniel EC (2023) Effective gene delivery using size dependant nano core-shell in human cervical cancer cell lines by magnetofection. PLoS ONE 18(9): e0289731. https://doi.org/10.1371/journal. pone.0289731

PROFESSIONAL DEVELOPMENT & TRAINING

Specialised Technical Workshop

- Flow Cytometry workshop, Department of Neurovirology, NIMHANS (2024)
- ELISA and PCR workshop for Congenital Rubella Syndrome, ICMR - NIV, Pune (2023)

AWARDS & RECOGNITION

- **COVID Warrior recognition, ICMR- National Institute of Virology- Pune (2020-2023)** Recognised for frontline service during COVID-19 pandemic response

- **Third best poster presentation award at the National Conference by National Corrosion Council of India. NCCI -2018**
  Title: '*Study of Biofilms in Drinking Water Distribution Systems Using a Novel Scientific tool: Foldscope*'.

TECHNICAL SKILLS

Molecular biology & Microbiology

- Quantitative PCR (qPCR) and RT-PCR, 16S rRNA gene sequencing
- ELISA, western blotting, MALDI-TOF mass spectrometry

Cell Biology

- Mammalian cell culture techniques, flow cytometry
- MTT assay, cell viability and proliferation assays

Advanced Instrumentation

- Next-generation sequencing - Ion Torrent and Illumina platforms
- Scanning Electron Microscopy (SEM), Atomic force Microscopy (AFM), Fluorescence Microscopy, Confocal Laser Scanning Microscopy (CLSM)
- UV-Visible spectroscopy, FT-IR spectroscopy and X-ray Diffraction

Data Analysis & Programming

- R and python programming for statistical analysis, data processing and visualisation